



BMS COLLEGE OF ENGINEERING

DEPARTMENT OF MACHINE LEARNING

LAB CYCLE – 2026 EVEN

VII Semester CIE

COURSE: MLOps	Date: 04th May 2026
COURSE CODE: 24AM6AEMLO	TOTAL: 30 Marks

Demonstrate a working solution on your laptop or on a lab system

Question Paper Code: mlops-cie-142	USN: 1BM23AI164
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This question paper is unique to you — do not share.

Instructions

1. You may use any **AI tools**, the internet, and any reference material
2. Use **random_state=42** and **test_size=0.2** for all train/test splits
3. Use MLflow for experiment tracking (local file store is fine)
4. All results must be **machine-readable JSON** — no screenshots
5. Do not modify the provided CSV data
6. The **results/*.json** files are your proof of execution — without them, you score 0 for that task

Submission

1. Create a GitHub repository named exactly: **Internals_Basics**
2. Inside it, create a folder named exactly: **MLOPs_Lab_CIE**
3. All your code, data, models, and results go inside MLOPs_Lab_CIE/
4. Make the repository **PUBLIC** before the deadline
5. Repo name and folder name must match **EXACTLY**

Scenario

You are an MLOps Engineer at **GeoSurvey**. GeoSurvey provides geotechnical assessment services. The risk team needs to predict land stability scores to guide construction planning and disaster preparedness.

Your dataset has these features:

- **slope_degrees** — (5–45)
- **rainfall_mm** — (50–500)
- **soil_depth_m** — (0.5–5)
- **vegetation_index** — (0.1–0.9)

Target variable: **land_stability_score** (score)

Tasks to be Performed

Complete these 4 tasks in sequence. Each builds on the previous one.

1. Experiment Tracking & Model Comparison (6 marks)
2. Docker Packaging (8 marks)
3. FastAPI Serving (8 marks)
4. Model Versioning (8 marks)

Expected Repository Structure

```
Internals_Basics/  
  MLOPs_Lab_CIE/
```

data/	
training_data.csv	<i>provided (do not modify)</i>
src/	
train.py	<i>training + MLflow tracking</i>
predict_cli.py	<i>CLI predictor</i>
api.py	<i>FastAPI service</i>
register_model.py	<i>model registration</i>
models/	<i>saved model files</i>
results/	
step1_s1.json	<i>Task 1 output</i>
step2_s3.json	<i>Task 2 output</i>
step3_s4.json	<i>Task 3 output</i>
step4_s6.json	<i>Task 4 output</i>
requirements.txt	
Dockerfile	<i>container definition</i>
.gitignore	

Dataset: training_data.csv

Copy the data below into **data/training_data.csv**:

```
slope_degrees,rainfall_mm,soil_depth_m,vegetation_index,land_stability_score
34.9,319.0,3.4,0.6,72.33
21.7,207.1,3.8,0.8,40.14
19.4,313.4,1.1,0.6,35.12
39.9,306.3,1.2,0.4,64.25
36.8,120.3,1.6,0.3,60.3
43.0,241.5,1.9,0.4,62.03
8.9,137.6,4.7,0.2,52.6
41.3,183.8,1.3,0.3,72.52
10.3,175.0,1.0,0.8,5.0
6.3,158.3,1.1,0.8,10.65
30.5,186.5,4.7,0.6,52.31
41.5,228.7,0.7,0.7,47.4
14.1,56.0,0.9,0.6,18.02
40.5,462.2,1.7,0.1,78.67
44.9,423.8,2.5,0.4,73.32
35.0,354.6,3.9,0.5,83.55
38.7,216.1,3.9,0.5,68.42
38.1,218.6,4.7,0.6,65.03
14.3,192.9,3.1,0.6,32.54
33.4,495.4,1.1,0.6,58.33
16.4,168.6,3.5,0.1,58.09
15.6,287.7,3.9,0.8,31.37
8.0,394.9,1.8,0.7,44.0
37.0,54.3,0.7,0.4,46.42
44.4,349.5,1.4,0.2,80.23
```

Task 1 — Experiment Tracking & Model Comparison (6 marks)

Train **SVR** and **RandomForest** models on the training data.

Log the following to MLflow:

- All model hyperparameters as params
- **MAE**, **RMSE** as metrics
- Tag: **priority = "high"**
- Experiment name: **"geosurvey-land-stability-score"**

Select the best model by **RMSE** (lower is better).

Save results to: **results/step1_s1.json**

```
{
  "experiment_name": "geosurvey-land-stability-score",
  "models": [
    {"name": "SVR", "mae": 0.0, "rmse": 0.0},
    {"name": "RandomForest", "mae": 0.0, "rmse": 0.0}
  ],
  "best_model": "...",
  "best_metric_name": "rmse",
}
```

```
    "best_metric_value": 0.0
}
```

Task 2 — Docker Packaging (8 marks)

Containerize a CLI prediction tool.

- Create **Dockerfile** with base image **python:3.12-slim**
- Create **src/predict_cli.py** that accepts features via argparse
- Build image as: **geosurvey-predictor:v1**

Test with:

```
docker run geosurvey-predictor:v1 --slope_degrees 31.9 --rainfall_mm 315.3 --soil_depth_m 2.4 --vegetation_index
```

Save results to: **results/step2_s3.json**

```
{
  "image_name": "geosurvey-predictor",
  "image_tag": "v1",
  "base_image": "python:3.12-slim",
  "test_input": {
    "slope_degrees": 31.9,
    "rainfall_mm": 315.3,
    "soil_depth_m": 2.4,
    "vegetation_index": 0.6
  },
  "prediction": 0.0
}
```

Task 3 — FastAPI Serving (8 marks)

Serve the best model from previous tasks via FastAPI.

- **POST /infer** — accepts features, returns prediction
- **GET /health** — returns {"status": "healthy", "model_loaded": true}
- Run on **port 8888**
- Pydantic validation for all input features with correct ranges
- Return HTTP 422 for invalid inputs

Test with this exact input:

```
{"slope_degrees": 31.9, "rainfall_mm": 315.3, "soil_depth_m": 2.4, "vegetation_index": 0.6}
```

Save results to: **results/step3_s4.json**

```
{
  "health_endpoint": "/health",
  "predict_endpoint": "/infer",
  "port": 8888,
  "health_response": {"status": "healthy", "model_loaded": true},
  "test_input": {"slope_degrees": 31.9, "rainfall_mm": 315.3, "soil_depth_m": 2.4, "vegetation_index": 0.6},
  "prediction": 0.0
}
```

Task 4 — Model Versioning (8 marks)

Register the model in the MLflow Model Registry.

- Registered model name: **"geosurvey-land-stability-score-predictor"**
- Retrieve and record the version number
- Link the registered version to the MLflow run

Save results to: **results/step4_s6.json**

```
{
  "registered_model_name": "geosurvey-land-stability-score-predictor",
  "version": 1,
  "run_id": "...",
  "source_metric": "rmse",
  "source_metric_value": 0.0
}
```